

Multiple Choice (10 marks)

1. Three of the following are classic security properties; which one is not?
 - (a) Confidentiality
 - (b) Availability
 - (c) Correctness
 - (d) Integrity
2. The Message confidentiality or privacy means that the sender and the receiver expect
 - (a) Integrity
 - (b) Confidentiality
 - (c) Authentication
 - (d) Nonrepudiation
3. What are the port states determined by Nmap?
 - (a) Active, inactive, standby
 - (b) Open, half-open, closed
 - (c) Open, filtered, unfiltered
 - (d) Active, closed, unused
4. Old operating systems like _____ and NT-based systems have buffer-overflow attack a common vulnerability.
 - (a) Windows 7
 - (b) Chrome
 - (c) IOS 12
 - (d) UNIX
5. _____ is the anticipation of unauthorized access or break to computers or data by means of wireless networks.
 - (a) Wireless access
 - (b) Wireless security
 - (c) Wired Security
 - (d) Wired device apps

True / False (10 marks)

Answer True or False

6. True or False: To verify a digital signature, the receiver needs the sender's public key.
7. True or False: Every cipher, including stream ciphers, can achieve perfect secrecy under certain conditions.
8. True or False: Penetration testing involves whole-system testing for security flaws and bugs to identify vulnerabilities.
9. True or False: Encapsulating Security Payload (ESP) is a component of the Secure IP Protocol used for securing internet protocol communications.
10. True or False: Message integrity ensures that data arrives at the receiver exactly as it was sent, without alterations during transmission.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to sort a list of lists by length and value.
12. Write a function to count the element frequency in the mixed nested tuple.

End of Paper